

WEEK 1 REPORT

DATA CLEANING AND TRANSFORMATION REPORT.

SUPERSTORE SALES DATA IMPORT, POWER QUERY & DATA CLEANING.

Introduction

*power BI is a tool used to analyze data and create reports. Power Query is used to clean and change the data before analysis.

*In this practical, I used the Superstore Sales Dataset in Power BI. I imported the CSV file and cleaned the data using Power Query. I checked the data types, missing values, spaces, and errors. I also created new date columns such as Year, Quarter, Month, Day, and Day of Week.

OBJECTIVES.

*To import the Superstore Sales dataset.

- *To understand Power BI views.
- *To check data types.
- *To find missing and error values.
- *To clean text data.
- *To rename columns.
- *To split columns when needed.
- *To create date columns.
- *To prepare clean data for analysis.

Tools and Dataset

Tool used: Microsoft Power BI Desktop.

preparation: Power Query Editor.

Dataset: Superstore Sales Dataset.

File Format: CSV.

1. Importing the Dataset .

The first step was to bring the Superstore Sales dataset into Power BI Desktop. Since the

dataset was available in CSV format, the Text/CSV option was used.

Steps Followed.

- *Opened Microsoft Power BI Desktop.
- *Selected the Home menu.
- *Clicked Get Data.
- *Selected Text/CSV.
- *Located the Superstore Sales CSV file.
- *Selected the required file and opened it.
- *Examined the preview of the dataset.
- *Selected Transform Data.
- *The data was opened in Power Query Editor.

Power Query Editor provided the required environment for inspecting and modifying the raw data.

samplesuperstore.csv

File Origin: 65001: Unicode (UTF-8) Delimiter: Comma Data Type Detection: Based on first 200 rows

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
1	US-2023-103800	1/3/2023	1/7/2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston
2	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
3	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
4	US-2023-112326	1/4/2023	1/8/2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville
5	US-2023-141817	1/5/2023	1/12/2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia
6	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
7	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
8	US-2023-106054	1/6/2023	1/7/2023	First Class	JO-15145	Jack O'Brian	Corporate	United States	Athens
9	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
10	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
11	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
12	US-2023-130813	1/6/2023	1/8/2023	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Angeles
13	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
14	US-2023-167199	1/6/2023	1/10/2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Hendersonville
15	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville
16	US-2023-105417	1/7/2023	1/12/2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville
17	US-2023-135405	1/9/2023	1/13/2023	Standard Class	M5-17830	Melanie Seite	Consumer	United States	Laredo
18	US-2023-135405	1/9/2023	1/13/2023	Standard Class	M5-17830	Melanie Seite	Consumer	United States	Laredo
19	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield
20	US-2023-149020	1/10/2023	1/15/2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield

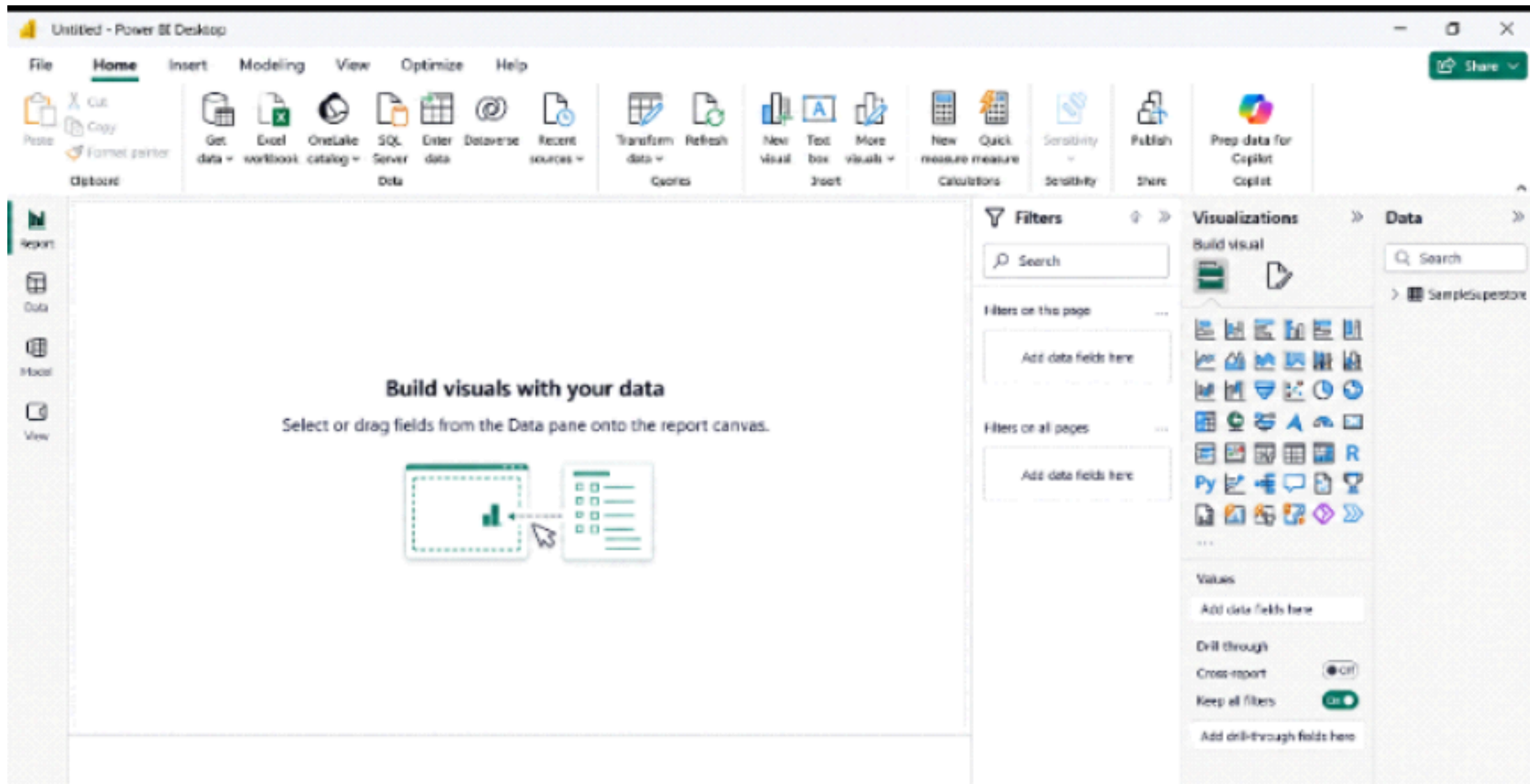
Extract Table Using Examples Load Transform Data Cancel

2.POWER BI VIEWS

Power BI Desktop provides different views that help in analyzing and managing data.

Report View

Report View is used to create reports and dashboards. It allows users to add visualizations such as charts, tables, cards, and graphs.



Data View

Data View displays the actual records contained in the tables. It is useful for checking rows, columns, and individual values.

samplesuperstore.csv

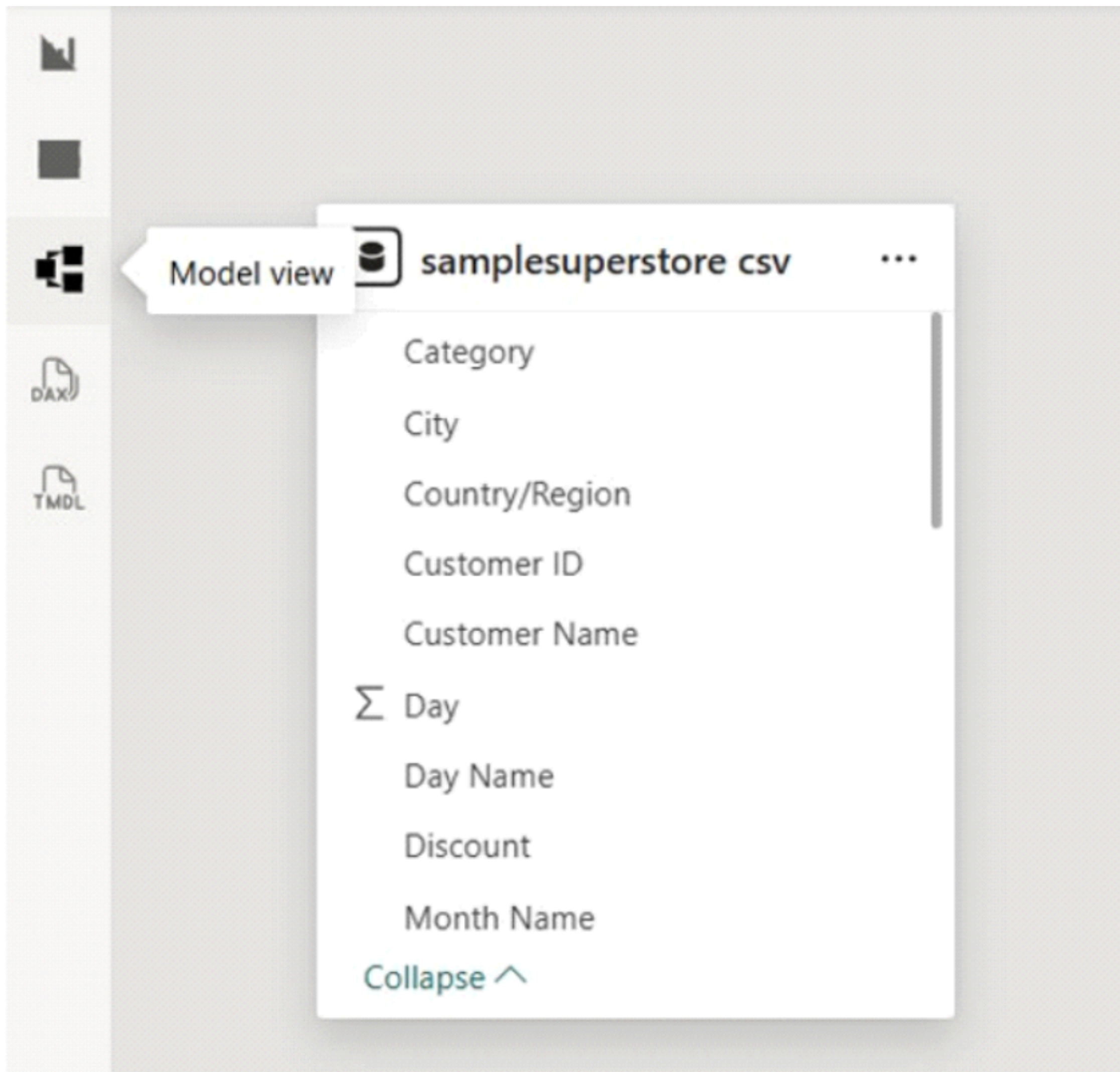
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Extract Table Using Examples | Load | Transform Data | Cancel

Model View

Model View provides a visual representation of tables and their relationships. It is useful when working with multiple tables.



Steps Followed

1. Opened the Power BI Desktop application.
2. Viewed the Report View.
3. Opened the Data View to inspect the dataset.

4. Checked the Model View.
 5. Reviewed the table structure and available fields.
 6. Confirmed that the dataset was loaded correct.
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3. Verifying and Modifying Data Types

After importing the dataset, I verified the data type assigned to each important column.

Correct data types are necessary to perform accurate calculations and analysis.

Steps Followed:

1. Opened the dataset in Power Query Editor.
2. Selected the required column.
3. Checked the data type displayed beside the column name.
4. Identified columns that required data

type changes.

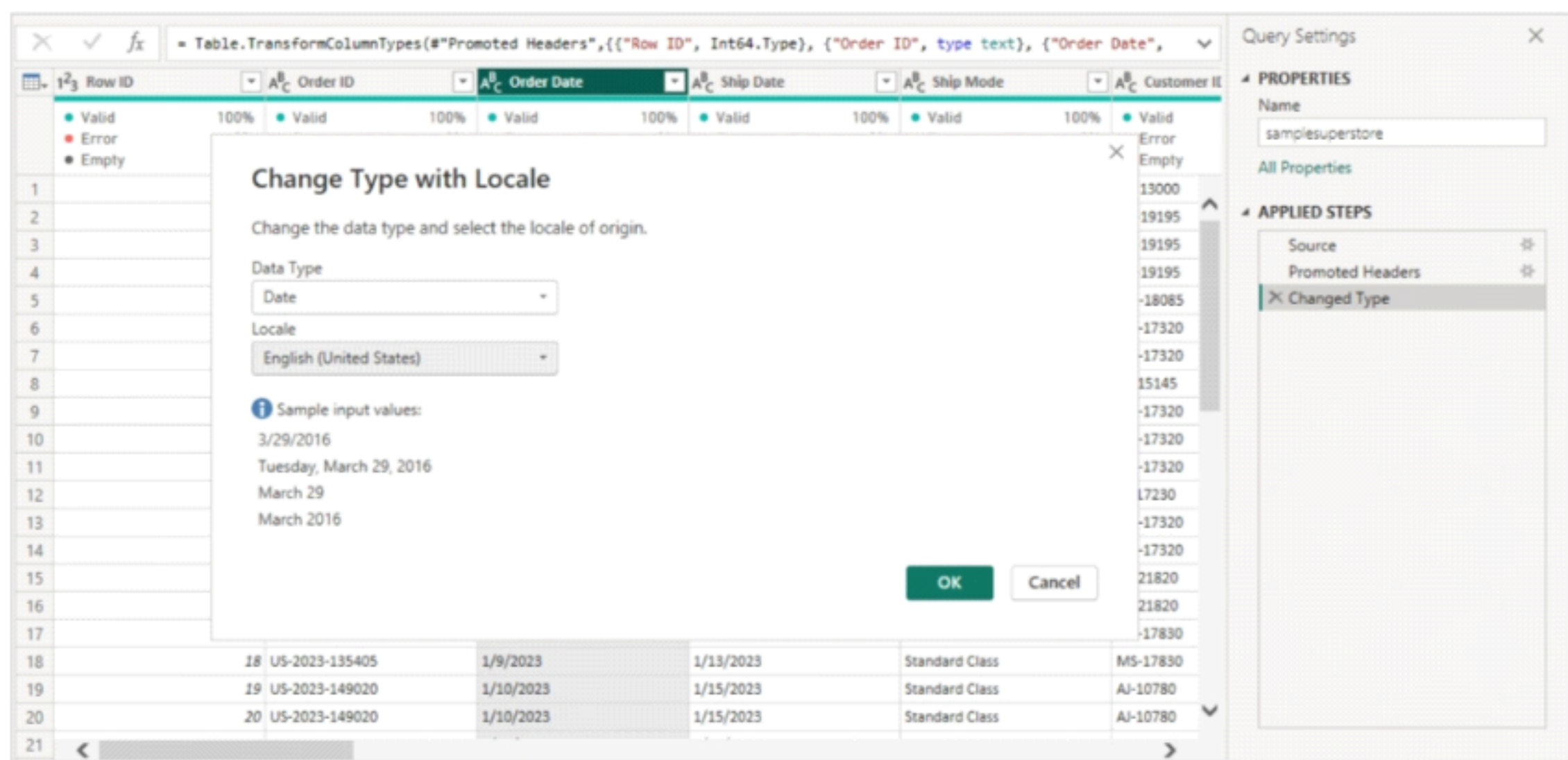
5. Used Data Type → Change Type to modify them.

6. Set categorical fields to Text.

7. Set quantity-related fields to Whole Number.

8. Set sales and profit fields to Decimal Number.

9. Changed the Order Date field to Date.



4.Cleaning Text Data Using Trim and Clean

Some text values may contain unwanted

spaces or characters. I used Trim and Clean to remove them.

The following columns were checked:

*Category

*Sub-Category

*Segment

Steps Followed:

1. Select the column.
2. Go to Transform.
3. Select Format.
4. Click Trim.
5. Again select Format.
6. Click Clean.
7. Check the cleaned values.

Product Name	Sales	Quantity
Message Book, Wirebound, Fo	16.448	
GBC Standard Plastic Binding S	3.54	
Avery 508	11.784	
SAFCO Boltless Steel Shelving	272.736	
Avery Hi-Liter EverBold Pen St	19.536	
Global Deluxe High-Back Mana	3573.83	
Rogers Handheld Barrel Pencil		
Dixon Prang Watercolor Pencil		
Ibico Hi-Tech Manual Binding S		
Alliance Super-Size Bands, Ass		
Southworth 25% Cotton Grant		
Xerox 225		
Wireless Extenders zBoost YX5		
GE 30524EE4	391.98	
Howard Miller 14-1/2" Diamet	76.728	
Acco Four Pocket Poly Ring Bir	10.43	
Newell 312	9.344	
Memorex Micro Travel Drive 8	31.2	
Howard Miller 11-1/2" Diamet	51.94	
Avery 482	2.89	
DAX Value U-Channel Document Frames, Easel Back	9.94	
O'Sullivan Elevations Bookcase, Cherry Finish	333.999	
Global Highback Leather Tilter in Burgundy	545.94	
4009 Highlighters by Sanford	19.9	
...	...	...

5. Checking Blank, Null and Error Values

The dataset was checked for missing or invalid values to ensure that the data was suitable for further analysis.

Steps Followed:

1. Selected the required column.
2. Opened the column filter.
3. Viewed the available values.
4. Checked for blank entries.
5. Looked for null values.
6. Checked for errors.
7. Reviewed the affected fields.
8. Corrected or handled values wherever required.
9. Rechecked the columns after processing.

*This step helped improve the consistency and reliability of the dataset.

6. Creating Month, Month Name and Quarter

After changing the Order Date to the correct Date type, I used the Order Date column to get useful date information. I created the required Month, Month

Name, and Quarter information for further analysis.

Steps Followed;

*Selected the **Order Date** column.

*Went to the **Add Column** tab.

*Used the available **Date** options.

7.Final Data Validation

After completing all the cleaning and transformation steps, I checked the dataset one more time. This was done to make sure that the changes were correct and the data was ready for further analysis.

Final Checks

- Checked the data types.
- Checked the Order Date column.
- Checked Month, Month Name, and Quarter columns.
- Checked blank and empty values.
- Checked null values.

- Checked error values.
- Checked the cleaned text data.
- Checked the renamed columns.
- Reviewed the final dataset.

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2	US-2013-112326	04-01-2023	08-01-2023	Standard Class	PO-19195
3	US-2013-112326	04-01-2023	08-01-2023	Standard Class	PO-19195
4	US-2013-112326	04-01-2023	08-01-2023	Standard Class	PO-19195
5	US-2013-141817	05-01-2023	12-01-2023	Standard Class	ME-18085
6	US-2013-167190	06-01-2023	10-01-2023	Standard Class	ME-17830
7	US-2013-167190	06-01-2023	10-01-2023	Standard Class	ME-17830
8	US-2013-106054	06-01-2023	07-01-2023	First Class	JD-15145
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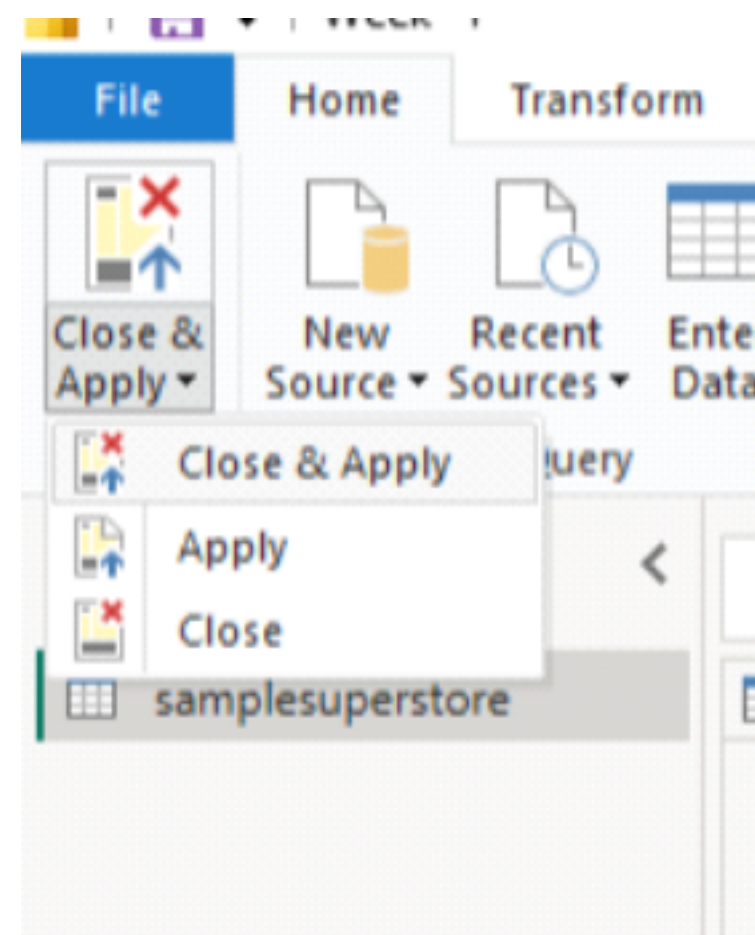
8.Close and Apply

After completing the cleaning and transformation process, I applied the changes to Power BI.

Steps Followed

- Checked the final dataset in Power Query Editor.
- Went to the **Home** tab.
- Clicked **Close & Apply**.

- Waited for Power BI to apply the changes.
- The cleaned dataset was loaded into Power BI.



RESULT

*The Superstore Sales dataset was successfully imported and preprocessed using Power BI and Power Query Editor.

*The dataset was examined for appropriate data types, missing values, blank values, null values, and errors. Text fields were standardized using Trim and Clean. Required columns were renamed and split where necessary.

*Date information was also transformed

to generate Year, Quarter, Month, Day, and Day of Week fields. Finally, the processed data was validated and loaded into the Power BI data model.

CONCLUSION

In this practical, I learned how to clean and transform data using Power BI and Power Query. I learned how to import a CSV file, check data types, remove unwanted spaces, find missing values, rename and split columns, and create date columns. This practical helped me understand the basic process of data cleaning and preparation before creating reports and dashboards.